

Discussion on “Joint Backtesting Procedure for Systemic Risk Measures”

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What the Paper Does

- The authors developed a comprehensive joint backtesting framework for systemic risk measures, e.g., MSE, CoVaR, and ΔCoVaR .
- The backtesting procedure is based on the forecast of the risk measure itself, rather than on the full specification of the risk model. → This allows the risk model to be flexible: parametric, nonparametric, ML, etc.
- A rigorous treatment for the estimation risk (parameter uncertainty) in the backtesting context is provided.

What the Paper Does

- Detailed Monte Carlo simulations are conducted to evaluate the finite-sample properties of the proposed joint backtesting.

Key findings:

- Standard backtesting approaches ignore estimation risk, and hence provide severely distorted test results.
- The distortion heightened with the ratio $p = N/M$. N : size of the out-of-sample observations; M : size of the in-sample observations.
- When the out-of-sample size is relatively larger, there will be more impact from estimation error on the test.
- The proposed test procedure is more robust on this. → Attain a violation rate that is within the acceptable bounds.

Some Comments

Check with different risk models:

- Confirm the claim in the paper that the proposed joint backtesting works with any risk model specifications.
- The distortion due to the estimation error might in fact be due to the model misspecification. → The risk model is restrictive & misspecified so that it generates inferior risk forecasts out-of-sample; fix the model instead of fixing backtesting?

Some Comments

Check with real data:

- Consider using stock (financial & banking sector) returns and the market returns in an empirical study. → Check whether the proposed approach evaluates risk models more correctly in practice.
- The sampled time period of returns should include different market conditions: calm & volatile market conditions, to test the robustness of the proposed method.

All the best for the future publication ^-^